Term 1 answer key

Formulas printed for students in every question. Answers with π are rounded to 1 decimal place, as each question says; probabilities are fractions in simplest form; every other answer is exact.

WEEK	SESSION	Q1 A Angles & shapes	Q2 B Measurement	Q3 C Number & percentages	Q4 D Algebra, ratio & rates	Q5 E Number plane, data & chance
1	1	152°	4 L	125	45 km/h	5/6
	2	115°	42 cm²	28 m	30 L per minute	7
	3	65°	31.4 cm	7/20	14	2/11
2	1	72°	60 cm²	−3	$x = 8$	164
	2	37°	90 cm²	$35	18 and 30	9
	3	133°	42 cm³	84%	$3x + 15$	(4, 1)
3	1	61°	52 m²	370	21:25	1/5
	2	48°	20 m	$35	16	1/5
	3	67°	9 cm	0.4	2 : 3	1/3
4	1	95°	37.7 cm	$6	$y = 6$	6
	2	150°	7000 m	−2	25 seconds	4
	3	122°	6 cm	12	Bag A ($3 per kg)	9
5	1	56°	14 cm	75%	$12 − 6y$	4/5
	2	66°	75 cm²	9	22	−4, 2
	3	125°	48 m	0.07	$20 and $40	7/10
6	1	115°	5 m	27 L	$x = 5$	7/12
	2	56°	64 cm²	0.82	12:45	15
	3	125°	44.0 cm	95%	$7y + 21$	8
7	1	110°	55 cm²	24	3 : 4	1/4
	2	66°	24 cm²	−12	8	23
	3	71°	25 000 cm²	4/5	60 km/h	1/6
8	1	60°	100 m	15	$m = 18$	(3, 5)
	2	130°	61 m²	9 kg	Bag B ($3 per kg)	17/20
	3	117°	196 cm²	35%	$4m + 36$	30
9	1	116°	75 m³	$175	24 and 32	2
	2	36°	78.5 cm	90%	35	21/50
	3	105°	91 m²	6.4	09:35	23
10	1	112°	56 cm²	−11	$x = 8$	1/3
	2	65°	52 cm	$36	2 : 3	8, 11
	3	140°	36 cm²	80%	$6x + 10$	12

A MA4-ANG-C-01 complementary and supplementary angles, angles at a point, vertically opposite angles, angles on parallel lines; MA4-GEO-C-01 angle sum of a triangle and a quadrilateral, exterior angle of a triangle

B MA4-LEN-C-01 perimeter, circumference, length units; MA4-ARE-C-01 area of rectangles, squares, triangles, parallelograms and composite shapes, area units; MA4-VOL-C-01 volume of rectangular prisms, volume and capacity units

C MA4-INT-C-01 order of operations with integers; MA4-IND-C-01 powers, square and cube roots; MA4-FRC-C-01 fraction and percentage of a quantity, one quantity as a percentage of another, fractions, decimals and percentages, multiplying and dividing by 10, 100 and 1000*

D MA4-ALG-C-01 substitution, the distributive law; MA4-EQU-C-01 one- and two-step equations; MA4-RAT-C-01 simplifying ratios, sharing in a ratio, rates, best buys, speed; 24-hour time and time differences*

E MA4-LIN-C-01 tables of values, reading points on the number plane; MA4-DAT-C-01 dot plots; MA4-DAT-C-02 mean, median, mode and range; MA4-PRO-C-01 probability, complementary events, experimental probability*

Codes are from the NSW Mathematics K–10 Syllabus (2022). The NESA website could not be reached from the build machine, so **none of these codes was confirmed on the NESA site**; they match the codes used in the Year 9 and Year 10 keys.

* the topic is in this outcome as far as could be checked, but where it sits in the syllabus (or, for 24-hour time, whether it is Stage 3 revision with no Stage 4 code) could not be confirmed.